

Design and Implementation of an Intelligent Microkernel OS using Linear Regression for Adaptive Power Management in IoT Edge Devices

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1. TITLE OF THE PROJECT

Design and Implementation of an Intelligent Microkernel OS using Linear Regression for Adaptive Power Management in IoT Edge Devices

2. INTRODUCTION / BACKGROUND

The rapid proliferation of Internet of Things (IoT) and Edge Computing has embedded smart sensor nodes into critical domains such as healthcare monitoring, precision agriculture, and industrial automation. Unlike traditional computing platforms, these edge devices operate on severely resource-constrained microcontrollers — typically with RAM measured in kilobytes — and rely on batteries that may power them for months or years without human intervention.

Lightweight Real-Time Operating Systems (RTOS) such as FreeRTOS and Zephyr OS are the preferred runtime environments for these devices. However, standard power-management strategies within such kernels are rule-based and static: the CPU is placed into a low-power sleep state based on fixed thresholds during idle periods. These methods fail to adapt to the dynamic and unpredictable nature of real-world IoT workloads, resulting in either unnecessary power dissipation when the CPU stays awake too long, or excessive wakeup latency when the system sleeps too deeply at the wrong moment.

This research bridges the gap between static OS scheduling and lightweight predictive intelligence. By embedding a Linear Regression model directly into the RTOS kernel idle hook — an area of the kernel specifically designed to execute code when no higher-priority task is active — the system evolves from a reactive architecture to a proactive, self-adapting power controller. This concept is aligned with the emerging field of Cognitive Operating Systems and the TinyML paradigm, which demonstrates that practical machine learning can run within severely memory-constrained environments.

3. STATEMENT OF THE PROBLEM

Traditional microkernel operating systems deployed in IoT environments rely on static, rule-based power management policies. These fixed threshold approaches present three critical limitations:

- They cannot model or predict the arrival time of future tasks based on changing workload patterns or environmental signals.

- The CPU either remains in an active high-power state for excessive durations (wasting battery capacity) or enters deep sleep blindly, generating significant and unpredictable wakeup latency when hardware interrupts occur.
- No existing lightweight, kernel-embedded predictive framework has been demonstrated to estimate idle durations accurately without creating unacceptable memory overhead or CPU cycle costs on microcontrollers.

This research directly targets this gap by proposing and validating a kernel-level Linear Regression predictor as a practical, deployable solution.

4. NEED AND RELEVANCE OF THE STUDY

As IoT networks scale to billions of globally distributed sensor nodes, manual battery maintenance becomes logistically impossible and financially prohibitive. Extending battery life through software-level intelligence — specifically at the OS kernel layer — offers scalable, hardware-independent benefits.

While powerful Deep Learning models achieve excellent scheduling predictions in cloud environments, their computational complexity ($O(n^2)$ or greater) and memory requirements (often megabytes of RAM) render them completely infeasible within a microkernel. This study proves that a simple mathematical model — Linear Regression ($y = mx + c$) — can deliver statistically meaningful predictions of CPU idle duration while consuming fewer than 2 KB of additional RAM and negligible CPU cycles. This validates TinyML not merely as an academic concept but as a practical kernel-level engineering strategy.

5. AIM AND OBJECTIVES OF THE STUDY

Aim: To design, implement, and empirically validate a self-adaptive microkernel power management module that uses an embedded Linear Regression model to predict CPU idle duration and thereby minimize energy consumption in simulated IoT edge workloads.

Objectives:

1. To analyse and modify the open-source FreeRTOS kernel's idle hook mechanism to extract and log real-time OS metrics including task queue depth, previous execution cycle durations, hardware interrupt frequencies, and idle interval histories.
2. To design, train, and embed a lightweight Linear Regression predictive engine into the FreeRTOS idle task sequence, enabling dynamic calculation of optimal CPU sleep durations based on historical execution patterns.
3. To simulate the modified intelligent RTOS under representative IoT workload scenarios on the QEMU emulator and conduct a rigorous comparative analysis against standard FreeRTOS regarding prediction accuracy (MSE), memory footprint overhead, wakeup latency, and estimated energy consumption.

6. RESEARCH QUESTIONS / HYPOTHESES

Research Questions:

- **RQ1:** Can a Linear Regression model operate reliably within FreeRTOS kernel space without causing memory overruns, scheduling interference, or degradation in real-time task responsiveness?
- **RQ2:** To what measurable extent does predictive sleep scheduling reduce total CPU active-state power consumption compared to the standard FreeRTOS tickless-idle mechanism, when evaluated under simulated periodic, sporadic, and burst IoT task distributions?
- **RQ3:** What is the quantified memory overhead (RAM + Flash) introduced by embedding the regression engine into the kernel, and does it remain within the 2 KB constraint suitable for ARM Cortex-M class microcontrollers?

Hypotheses:

- **H1 (Primary Hypothesis):** Embedding a trained Linear Regression model into the FreeRTOS idle hook will reduce simulated microcontroller energy consumption by at least 20% compared to static tickless-idle, while maintaining a total memory overhead of less than 2 KB.
- **H2 (Secondary Hypothesis):** The regression-based predictor will achieve a Mean Squared Error (MSE) of less than 5% on idle duration prediction across mixed IoT workload profiles.

7. LITERATURE REVIEW (BRIEF)

The energy optimization challenge in RTOS-based embedded systems has been addressed through several generations of techniques. Early approaches introduced Tickless Idle Mode — a mechanism that stops the periodic OS timer tick during idle periods to eliminate unnecessary interrupt overhead [7]. While effective, this approach relies solely on real-time detection rather than anticipation of future task arrivals.

The emergence of TinyML has demonstrated that machine learning inference can execute on sub-milliwatt microcontrollers [3]. However, most documented implementations focus on application-layer use cases such as keyword detection or gesture recognition — not OS-level scheduling. Reinforcement Learning (Q-learning) has been explored for CPU scheduling on Linux environments [6], but its $O(n^2)$ complexity disqualifies it for microcontrollers with 16–64 KB RAM.

Lightweight regression techniques have been shown to require negligible CPU cycles for inference [4], and time-series prediction using regression models has demonstrated strong performance in resource allocation problems at the hardware edge [9]. However, no prior study has directly mapped RTOS kernel idle-task metrics into a regression matrix embedded within kernel space itself. This research fills that specific and critical gap, contributing a novel proof-of-concept for Kernel-Space TinyML in FreeRTOS.

8. RESEARCH METHODOLOGY

Research Design: Experimental and Quantitative. A controlled baseline FreeRTOS environment will be established, systematically modified with the regression module, and tested under defined simulated workloads with all variables controlled and measured.

Data Sources: Secondary datasets generated through real-time internal logging of FreeRTOS task execution timelines, idle period durations, hardware interrupt frequencies, and CPU utilization patterns via software instrumentation hooks inserted into the kernel source code.

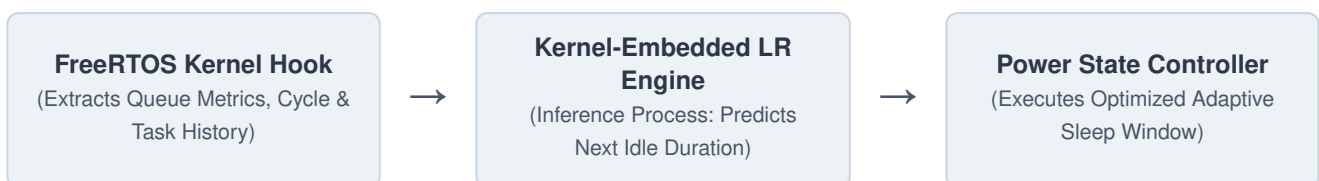
Simulation Environment: QEMU Emulator configured to emulate an ARM Cortex-M4 class processor. This provides a reproducible, deterministic execution environment without requiring physical hardware, enabling accurate cycle-level measurement.

Workload Profiles: Three representative IoT task distribution patterns will be tested: (i) Periodic Sensor Sampling — fixed-interval tasks typical of temperature or pressure sensors; (ii) Sporadic Event-Driven — irregular interrupt-driven tasks typical of alarm systems; (iii) Burst Transmission — clustered high-frequency tasks typical of wireless data upload cycles.

Sample Size: 10,000 continuous simulated task execution cycles per workload profile, yielding a statistically robust dataset for regression training and evaluation.

Data Analysis Tools: Python (Pandas, NumPy, Matplotlib, Scikit-learn) for regression model training, MSE calculation, memory profiling, and performance graph generation. GCC compiler map files for precise Flash and RAM footprint quantification.

System Architecture Pipeline:



9. EXPECTED OUTCOMES / DELIVERABLES

- **A Modified FreeRTOS Kernel Source Code** repository containing the fully integrated Linear Regression idle-hook predictive scheduling module, documented and version-controlled.
- **A QEMU Simulation Sandbox** configured for ARM Cortex-M4 emulation, capable of reproducing all experimental workload scenarios and logging energy-proxy metrics.
- **Comparative Performance Analysis Graphs** quantifying energy saved (active CPU time reduction %), prediction accuracy (MSE), wakeup latency overhead, and memory footprint (RAM + Flash) versus standard FreeRTOS across all three workload profiles.

- **An MSc Computer Science Research Thesis** presenting the complete mathematical derivation, kernel implementation, experimental methodology, statistical results, and conclusions regarding the viability of Kernel-Space TinyML for IoT power optimization.

10. REFERENCES / BIBLIOGRAPHY (IEEE STYLE)

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